



YouTube link: YouTube. <https://www.youtube.com/watch?v=JLiF7gZ2rjo>

Individual Project (Flymandu)

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ST5007CEM Web Development

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Contents

Introduction	4
Aim	5
Objectives	5
Learning Outcomes	6
Features Included in the system	7
User Login for Admin and User:	7
User Registration:	8
Homepage:	9
Booking Page:	11
Ticket-Status Page:	13
Flight-Routes Page:	15
Admin-Dashboard Page:	16
Add-flight Page:	17
View-Flights Page:	18
View-Users Page:	19
Pending bookings Page:	20
Tools and Technologies	21
ER Diagram	22
Use-case Diagram	23
Database Overview	24
Testing	25
Future features	27
Conclusion	28
Reference	29

Figure 1LoginPage	8
Figure 2SignupPage.....	8
Figure 3Homepage1.....	10
Figure 4Homepage2.....	10
Figure 5Booking page.....	12
Figure 6Ticket-Status	14
Figure 7Flight-Routes.....	15
Figure 8AdminDashboard	16
Figure 9AddFlight-page.....	17
Figure 10ViewFlights.....	18
Figure 11View-Users	19
Figure 12Pending-flights	20
Figure 13ToolsNTech.....	21
Figure 14Er diagram.....	22
Figure 15UseCase.....	23
Figure 16Backend-Flow.....	24
Figure 17UnitTest1.....	25
Figure 18UnitTest2.....	26
Figure 19BDDtest	26

Introduction

Flymandu is a state-of-the-art online platform designed to simplify the process of booking flights with various airlines. With a focus on intuitive user experience and seamless integration, Flymandu provides a comprehensive solution for both travelers and airlines. The platform, built using React for the frontend and a robust Spring Boot backend, ensures a smooth and secure booking experience for all users.

Flymandu offers a range of features to enhance the travel booking process. Users can easily search for flights, view detailed information, and make bookings with a few simple clicks. The platform supports a variety of airlines, providing users with numerous options to suit their travel needs. For administrators, Flymandu provides tools to manage and update flight information effortlessly.

One of the standout features of Flymandu is its streamlined booking management system, which allows users to track and manage their bookings with ease. The platform also includes an integrated notifications system, keeping users informed about their booking status and any important updates.

Flymandu is dedicated to delivering a user-centric experience. The platform promotes efficient flight management and offers a seamless interface for booking and tracking flights. Whether you are planning your next trip or managing flight details, Flymandu provides a reliable and enriching experience for all travelers.

Aim

To create a dynamic and user-friendly platform for booking flights, connecting travelers with a diverse range of airlines. By offering a streamlined booking experience, robust flight management tools, and interactive features, Flymandu aims to enhance the convenience and efficiency of air travel. Our goal is to foster a community of informed travelers who can easily search for, book, and manage their flights, while benefiting from an engaging and supportive travel ecosystem.

Objectives

- **Seamless Flight Search and Booking:** Provide a user-friendly interface for searching and booking flights across various airlines, ensuring a smooth and efficient booking experience.
- **Comprehensive Flight Management:** Enable users to view, manage, and track their flight bookings with ease, including options to modify or cancel reservations as needed.
- **Admin Tools:** Equip administrators with robust tools for managing flight information, approving bookings, and overseeing the overall operation of the platform.
- **User Profile Management:** Allow users to create and manage profiles, storing relevant personal and travel information for a more personalized experience.
- **Enhanced Security:** Ensure the security and privacy of user data and transactions through robust backend protection and secure data handling practices.
- **Responsive Design:** Develop a responsive design that provides a consistent and user-friendly experience across various devices and screen sizes.
- **Community Engagement:** Foster a community of travelers by providing features that encourage interaction, feedback, and sharing of travel experiences.

Learning Outcomes

By working on the Flymandu project, you will achieve several key learning outcomes.

You'll gain valuable experience in enhancing user experience design through React, focusing on creating intuitive and engaging interfaces. You'll also develop advanced backend skills using Spring Boot, including managing API endpoints, user authentication, and data handling.

Implementing interactive features, such as chatbots and real-time notifications, will deepen your understanding of enhancing user engagement and providing personalized support.

Additionally, you'll learn about the complexities of flight management systems, from booking processes to administrative tools. You'll acquire knowledge in implementing security best practices to protect user data and ensure secure transactions. Mastering responsive design principles will enable you to create a seamless experience across various devices.

Experience in database management will be gained by designing and managing systems for efficient storage of flight, booking, and user data. You'll enhance your problem-solving and debugging skills by addressing development challenges and integrating APIs. The project will also develop your project management abilities, including version control and collaboration.

Lastly, incorporating user feedback into the development process will help you continually improve the platform and adapt to user needs, preparing you for future projects and professional opportunities.

Features Included in the system

User Login for Admin and User:

Description: The Flymandu platform features a dual login system that caters to two primary user roles: Admins and Users. This separation ensures that each role has access to specific functionalities and features tailored to their needs.

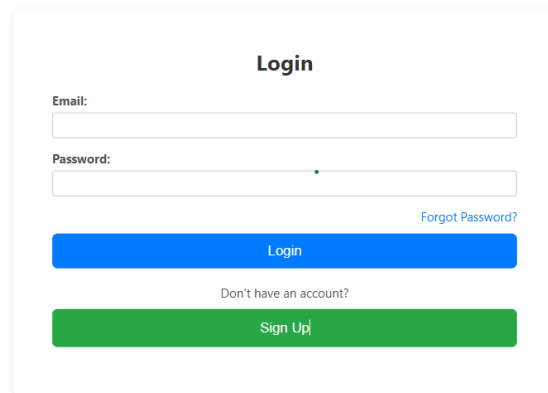
- **User Login:**

Purpose: Allows regular users to log in and access features such as flight search, booking management, and personal account settings. Access Rights: Users can search for flights, make bookings, view and manage their reservations, and receive real-time notifications about their bookings.

- **Admin Login:**

Purpose: Provides administrators with access to the backend of the platform, where they can manage flight information, oversee booking approvals, and perform administrative tasks. Access Rights: Admins can add, update, and delete flight listings, approve or reject bookings, and view comprehensive reports on user activity and system performance.

This dual login system ensures that each user type has access to the appropriate tools and features, enhancing both security and functionality by restricting access based on user role.

The image shows a login form titled "Login" in bold black text. Below the title, there are two input fields: "Email:" and "Password:". The "Password:" field has a small green dot at the end. To the right of the "Password:" field is a link that says "Forgot Password?". Below these fields is a blue button with the text "Login". Below the button is a link that says "Don't have an account?". At the bottom is a green button with the text "Sign Up".

Login

Email:

Password:

[Forgot Password?](#)

Login

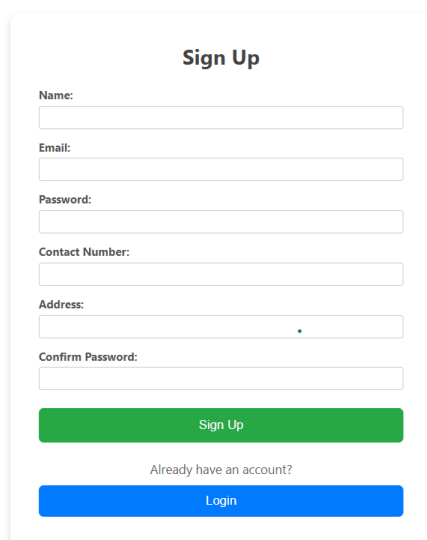
[Don't have an account?](#)

Sign Up

Figure 1LoginPage

User Registration:

The Flymandu platform includes a comprehensive user registration system that allows new users to create accounts and access personalized features based on their roles. The registration process is designed to be simple and secure, ensuring a smooth onboarding experience for both general users and administrators.

The image shows a sign up form titled "Sign Up" in bold black text. Below the title, there are six input fields: "Name:", "Email:", "Password:", "Contact Number:", "Address:", and "Confirm Password:". The "Address:" field has a small green dot at the end. Below these fields is a green button with the text "Sign Up". Below the button is a link that says "Already have an account?". At the bottom is a blue button with the text "Login".

Sign Up

Name:

Email:

Password:

Contact Number:

Address:

Confirm Password:

Sign Up

[Already have an account?](#)

Login

Figure 2SignupPage

Homepage:

The Flymandu homepage is designed to provide users with an engaging and intuitive introduction to the platform. At the top of the page, the navbar is prominently displayed, offering a fixed and accessible menu that includes links to key sections such as Home, Flights, My Bookings, Profile, and Help. For admins, a dedicated link to the Admin Dashboard is also available. The navbar features the Flymandu logo, which serves as a clickable link back to the homepage, and includes a login/logout button that adapts based on user authentication status.

The introduction section on the homepage provides a warm welcome to Flymandu, highlighting the platform's primary features and benefits. It includes an interactive search bar that enables users to begin their flight search immediately. This section briefly explains how users can search for flights, manage their bookings, and utilize the integrated chatbot for personalized assistance. Additionally, it offers insights into community engagement opportunities, such as providing feedback and exploring user interactions.

At the bottom of the page, the footer ensures users have easy access to important information and resources. It contains contact details, links to the privacy policy, terms of service, and an about us section, along with social media icons for connecting with Flymandu on various platforms. A newsletter signup option is also available for users to receive updates and news about the platform. The footer is fixed at the bottom of the page, maintaining its visibility and accessibility across different sections of the website.

Overall, the homepage is designed with a responsive layout to provide a consistent experience across devices, featuring a visually appealing design that includes high-quality images, clean typography, and a cohesive color scheme. Accessibility features ensure that all users can navigate the site effectively, making the Flymandu homepage both functional and inviting.

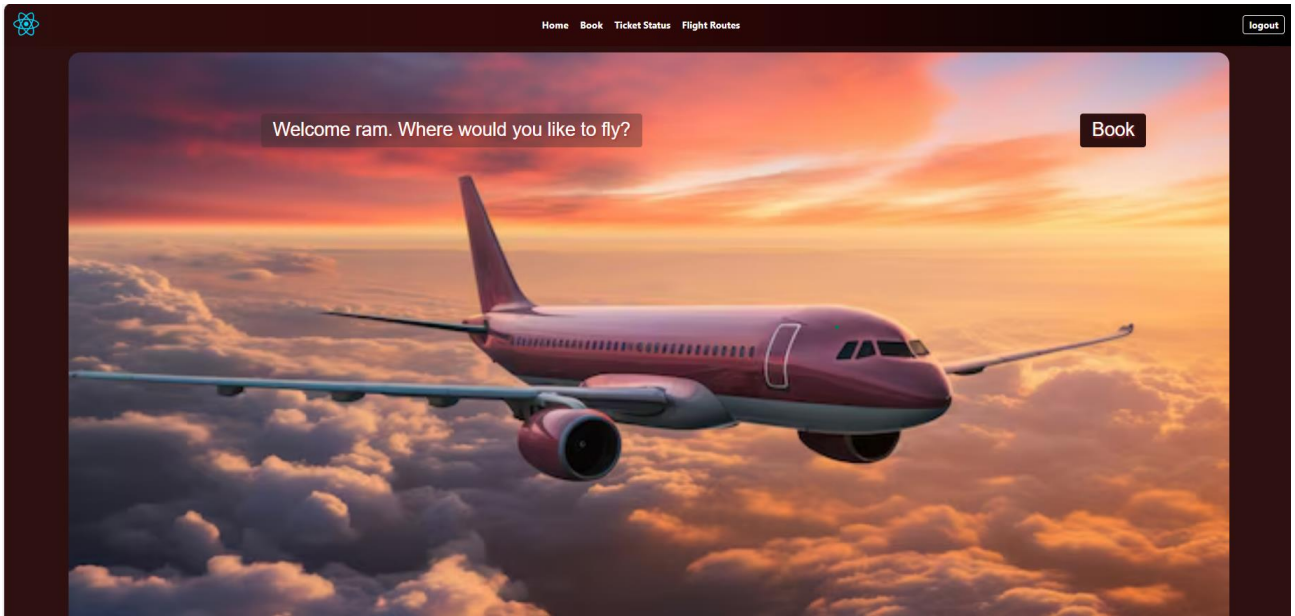


Figure 3Homepage1

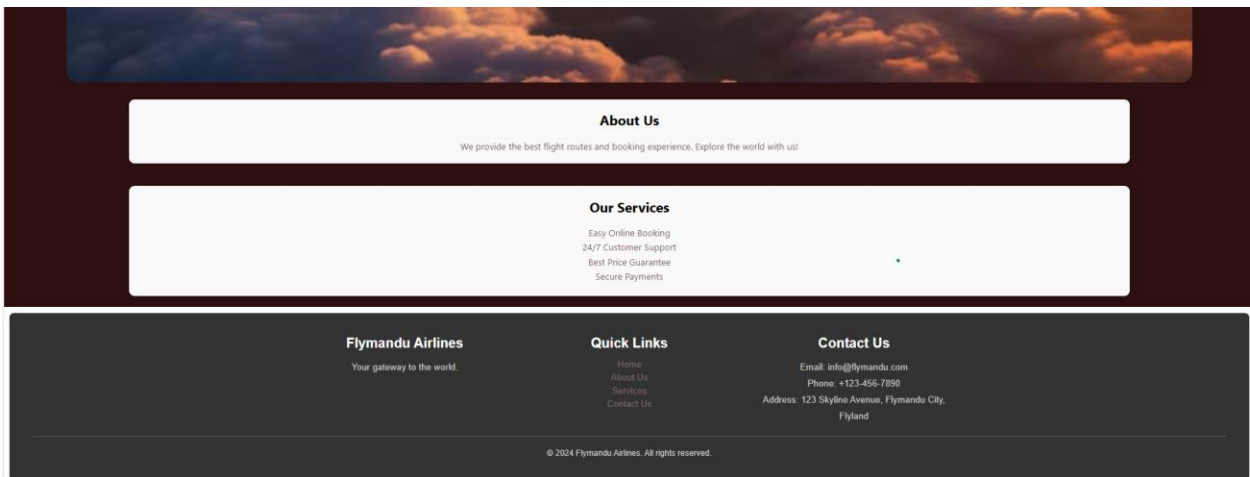


Figure 4Homepage2

Booking Page:

The Booking Page on Flymandu is designed to offer users a seamless and intuitive flight search experience. At the heart of this page is a comprehensive list of flights, which showcases various travel options based on the user's search criteria. Each flight listing provides essential details, such as departure and arrival times, airline information, and fare details, enabling users to compare different flights and select the one that best suits their needs.

To enhance the search experience, the Booking Page features a filter panel positioned on the side. This panel includes two key filters: the Airline Filter and the Price Filter. The Airline Filter allows users to select specific airlines from a list, helping them narrow down their search results to their preferred carriers. Meanwhile, the Price Filter offers a range slider or input fields where users can set their desired price range. This feature helps users find flights within their budget by adjusting the search results according to their specified price limits.

The layout of the Booking Page is designed to be both clean and user-friendly. The flight listings are prominently displayed in the center of the page, making it easy for users to browse through the available options. The filter panel is conveniently located to the side, allowing users to apply their search preferences without cluttering the main flight listings area. This thoughtful design ensures that users can efficiently find and book flights that meet their specific requirements, providing a streamlined and enjoyable booking experience.

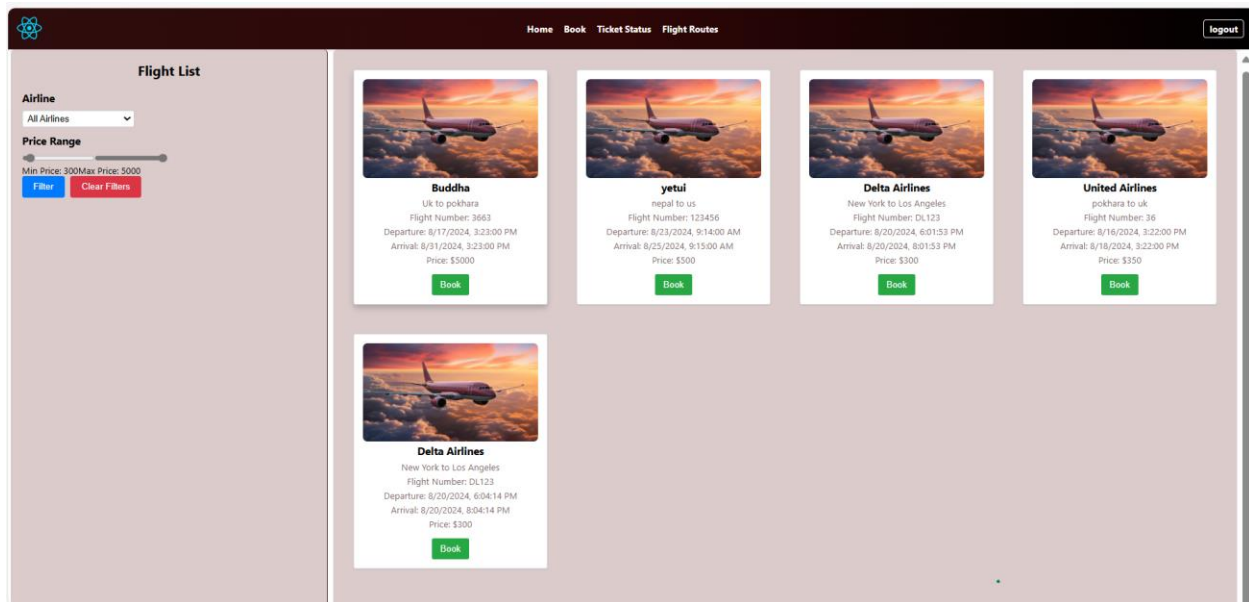


Figure 5Booking page

Ticket-Status Page:

The Ticket-Status Page on Flymandu provides users with a detailed overview of their booking statuses, enhancing their ability to manage and track their reservations efficiently. This page displays a list of all booked flights, categorized by their current status, which can be approved, declined, or pending.

For each booking, users can view its status at a glance, allowing them to quickly determine whether their flight has been confirmed, rejected, or is still awaiting approval. In addition to status information, the page includes a notable feature for approved bookings: users have the option to download a PDF of their ticket. This functionality ensures that users can easily access and store a copy of their flight details for their records.

The Ticket-Status Page also includes an option to cancel bookings. Users can select a booking and initiate the cancellation process directly from the page. This feature provides flexibility and control, allowing users to manage their reservations with ease.

Overall, the Ticket-Status Page is designed to offer a comprehensive view of booking statuses, provide convenient access to important ticket information, and facilitate the cancellation process, ensuring a smooth and user-friendly experience for managing flight reservations.

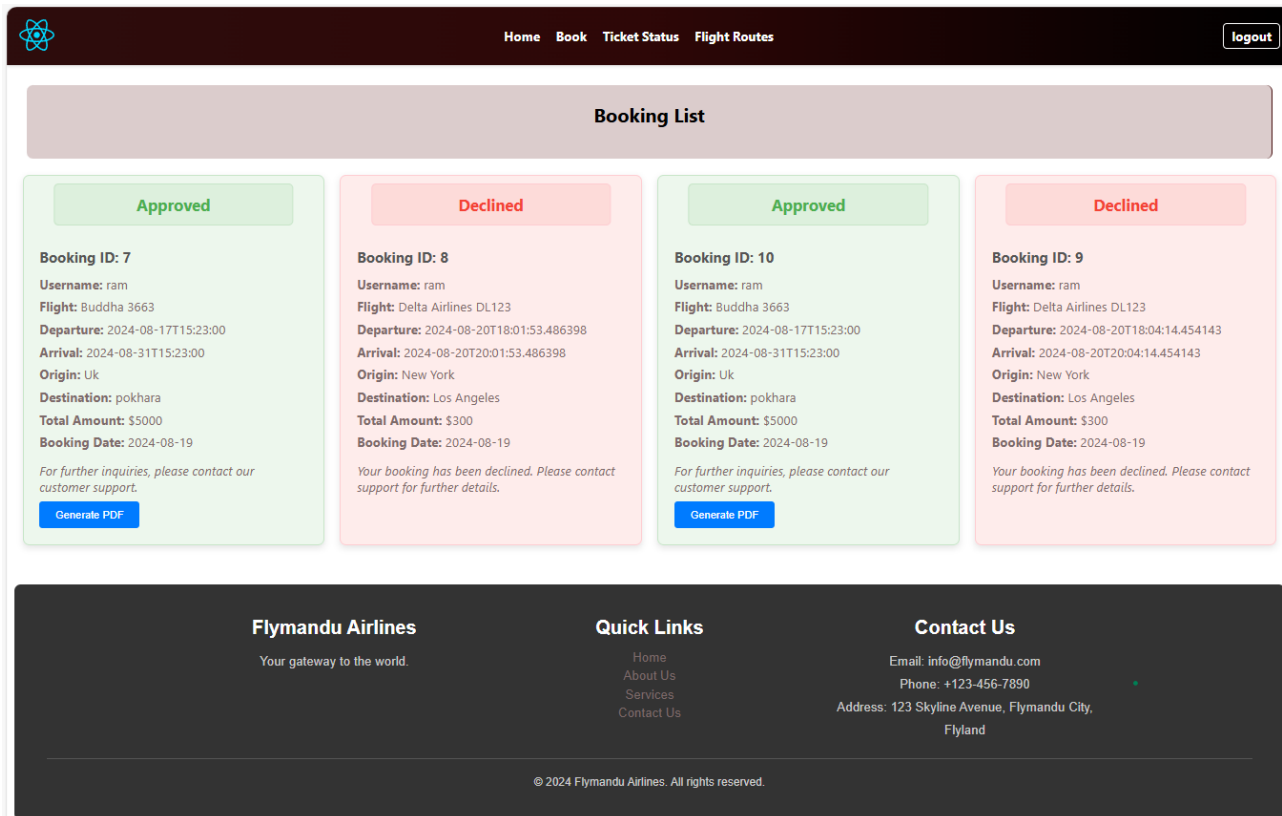


Figure 6Ticket-Status

Flight-Routes Page:

The Flight-Routes Page on Flymandu provides users with a powerful tool to find and book flights tailored to their travel needs. This page features a set of filters that enable users to refine their search based on various criteria such as origin, destination, and travel dates. By entering their desired starting point and destination, as well as specifying their travel dates, users can easily narrow down the available flight options to match their preferences.

Once the filters are applied, the page presents a list of flights that meet the selected criteria, displaying essential details such as departure and arrival times, airline information, and fare options. This allows users to compare different flights and make informed decisions.

Additionally, the Flight-Routes Page includes a convenient booking option directly within the flight listings. Users can proceed with their booking seamlessly from this page, ensuring a streamlined and efficient process for securing their travel arrangements.

Overall, the Flight-Routes Page is designed to enhance the user experience by providing an intuitive interface for filtering flights and booking directly from the search results, making it easier for users to find and book the flights that best suit their travel plans.

The screenshot displays the Flymandu website's Flight-Routes page. At the top, a dark navigation bar contains links for Home, Book, Ticket Status, and Flight Routes, along with a login button. Below the navigation bar is a search form titled "Search Flights" with input fields for Origin (pre-filled with "new york") and Destination, and a blue "Search" button. Below the search form, two flight listings are shown side-by-side. Each listing includes the airline (Delta Airlines), flight number (DL123), departure and arrival times, origin (New York), destination (Los Angeles), and price (\$300), with a green "Book" button at the bottom. The footer section is dark and contains three columns: "Flymandu Airlines" with the tagline "Your gateway to the world", "Quick Links" with links to Home, About Us, Services, and Contact Us, and "Contact Us" with email, phone, and address information. A copyright notice "© 2024 Flymandu Airlines. All rights reserved." is at the bottom.

Flight Details	Flight Details
Airline: Delta Airlines	Airline: Delta Airlines
Flight Number: DL123	Flight Number: DL123
Departure Time: 8/20/2024, 8:01:53 PM	Departure Time: 8/20/2024, 8:04:14 PM
Arrival Time: 8/20/2024, 8:01:53 PM	Arrival Time: 8/20/2024, 8:04:14 PM
Origin: New York	Origin: New York
Destination: Los Angeles	Destination: Los Angeles
Price: \$300	Price: \$300
Book	Book

Flymandu Airlines

Your gateway to the world.

Quick Links

- [Home](#)
- [About Us](#)
- [Services](#)
- [Contact Us](#)

Contact Us

Email: info@flymandu.com

Phone: +123-456-7890

Address: 123 Skyline Avenue, Flymandu City, Flyland

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Figure 7Flight-Routes

Admin-Dashboard Page:

The Admin-Dashboard Page on Flymandu offers a robust and intuitive interface for administrators to efficiently manage the platform. The page is equipped with a sidebar that facilitates easy navigation through key administrative functions. The sidebar includes options to Add Flights, enabling admins to input new flight information; View Flights, which displays a comprehensive list of all current flights for monitoring and management; View Pending Flights, allowing admins to review and approve flights awaiting approval; and View Users, where admins can manage and review user accounts.

At the top of the page, the navbar provides a straightforward logout option, ensuring that admins can securely end their session. The footer at the bottom of the page includes essential links such as contact information, privacy policy, and terms of service, providing additional resources and support. Overall, the Admin-Dashboard Page is designed to deliver a seamless administrative experience, combining intuitive navigation with essential management tools to oversee and control various aspects of the Flymandu platform.

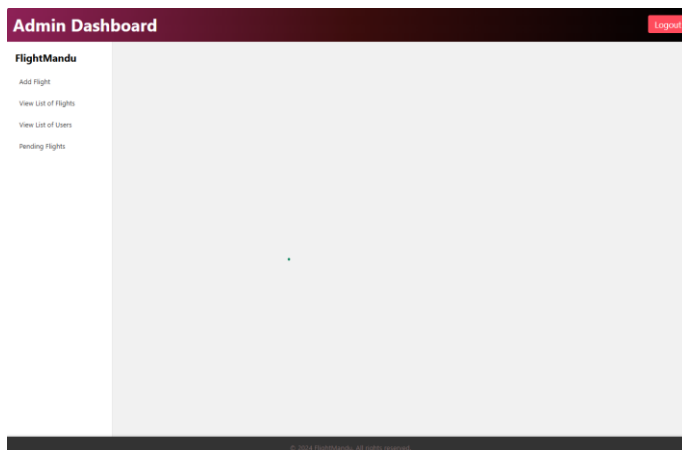


Figure 8AdminDashboard

Add-flight Page:

The Add-Flight Page on Flymandu is specifically designed to simplify the process of adding new flight information to the platform. This page features a straightforward form that administrators can use to input essential details about a flight. The form includes fields for crucial information such as the flight number, airline, departure and arrival cities, departure and arrival dates and times, and the flight price. By filling out these fields, administrators can easily update the flights table with new flight options. Once the form is completed and submitted, the new flight information is added to the system, ensuring that users have access to the most current flight options for booking. The page is designed for ease of use, making it simple for admins to manage and expand the flight offerings on Flymandu.

Admin Dashboard [Logout](#)

FlightMandu

- Add Flight
- View List of Flights
- View List of Users
- Pending Flights

Add Flight

Airline:

Flight Number:

Departure Time:

Destination:

Origin:

Arrival Time:

Price:

[Add Flight](#)

Figure 9AddFlight-page

View-Flights Page:

The View-Flights Page on Flymandu provides administrators with a comprehensive overview of all the flights that have been added to the platform. This page displays a detailed list of flights, including key information such as flight numbers, airlines, departure and arrival cities, dates and times, and prices.

In addition to viewing the list of flights, administrators have the capability to update or remove flight entries directly from this page. To update a flight, admins can select the flight they wish to modify, make necessary changes to the flight details, and then save the updated information.

This ensures that the flight information remains accurate and current.

For removing a flight, admins can choose to delete specific flight entries from the list. This feature allows for efficient management of the flight database by enabling the removal of outdated or incorrect flight options.

Overall, the View-Flights Page is designed to provide a user-friendly interface for managing flight information, offering both visibility and control over the flights available on Flymandu.

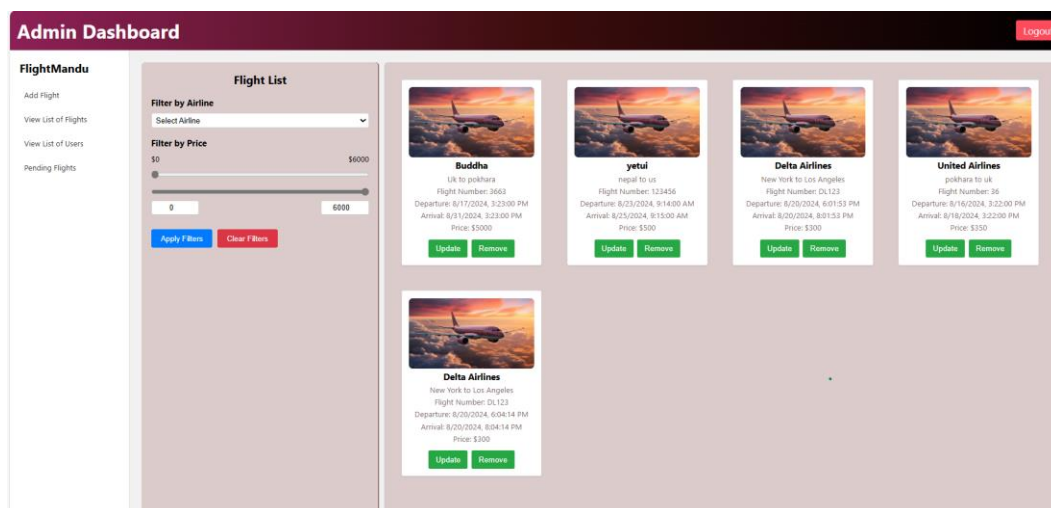


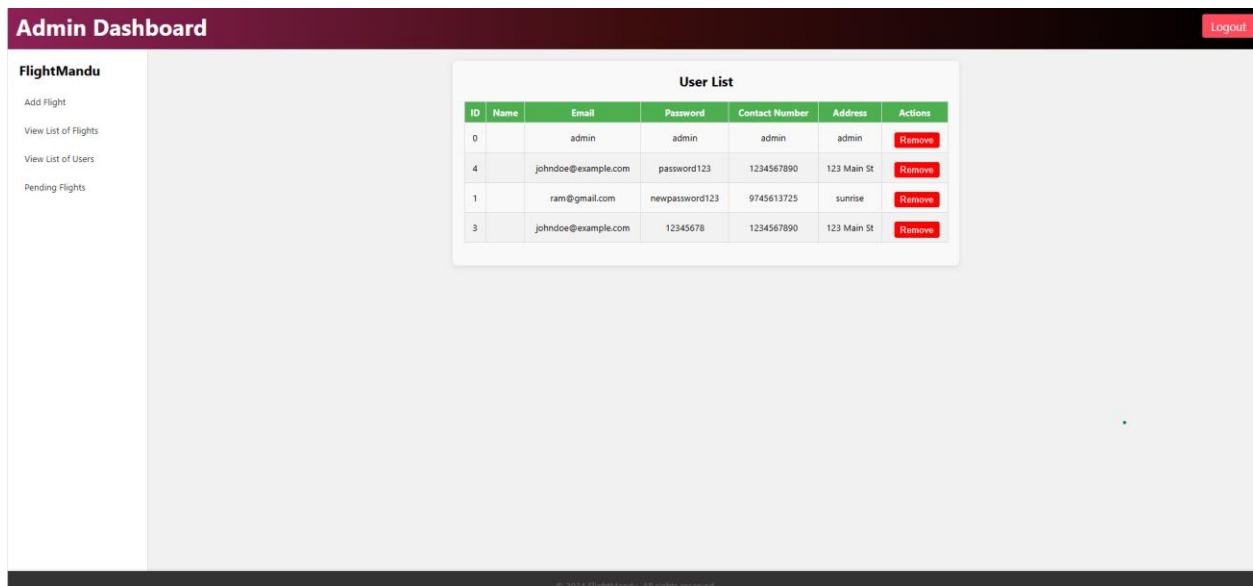
Figure 10ViewFlights

View-Users Page:

The View-Users Page on Flymandu provides administrators with a detailed overview of all existing users in the system. This page displays a list of users, including key information such as user IDs, names, email addresses, and roles. This comprehensive view allows admins to monitor and manage user accounts effectively.

Administrators have the ability to remove users directly from this page if necessary. By selecting a user from the list, admins can initiate the removal process, which deletes the user's account from the system. This feature ensures that admins can manage user access and maintain the integrity of the platform by removing inactive or problematic accounts.

Overall, the View-Users Page is designed to offer a clear and efficient way for administrators to oversee user accounts, providing both visibility into user information and control over account management.



The screenshot displays the Admin Dashboard for FlightMandu. The dashboard has a dark purple header with the title 'Admin Dashboard' and a 'Logout' button. On the left, there is a sidebar with the 'FlightMandu' logo and a list of navigation links: 'Add Flight', 'View List of Flights', 'View List of Users', and 'Pending Flights'. The main content area features a 'User List' table with the following data:

ID	Name	Email	Password	Contact Number	Address	Actions
0		admin	admin	admin	admin	Remove
4		john.doe@example.com	password123	1234567890	123 Main St	Remove
1		ram@gmail.com	newpassword123	9745613725	sunrise	Remove
3		john.doe@example.com	12345678	1234567890	123 Main St	Remove

At the bottom of the dashboard, there is a footer that reads '© 2024 FlightMandu. All rights reserved.'

Figure 11View-Users

Pending bookings Page:

The Pending-Bookings Page on Flymandu is designed to facilitate the management of flight bookings that are awaiting administrative review. This page displays a comprehensive list of all booked flights that have not yet been processed, providing administrators with a clear overview of pending requests. Each booking entry includes detailed information such as the booking reference, user details, flight specifics, and the current status of the booking.

Administrators can take action on each booking by either approving or declining it. If a booking is in order and meets all necessary criteria, the admin can approve it, which updates the booking status to approved and confirms the flight for the user. Conversely, if a booking cannot be processed, the admin has the option to decline it, updating the status to declined and typically notifying the user of the reason for the decline.

This page is designed to streamline the process of handling pending bookings, allowing administrators to efficiently manage and review each request while maintaining the integrity and efficiency of the booking system.

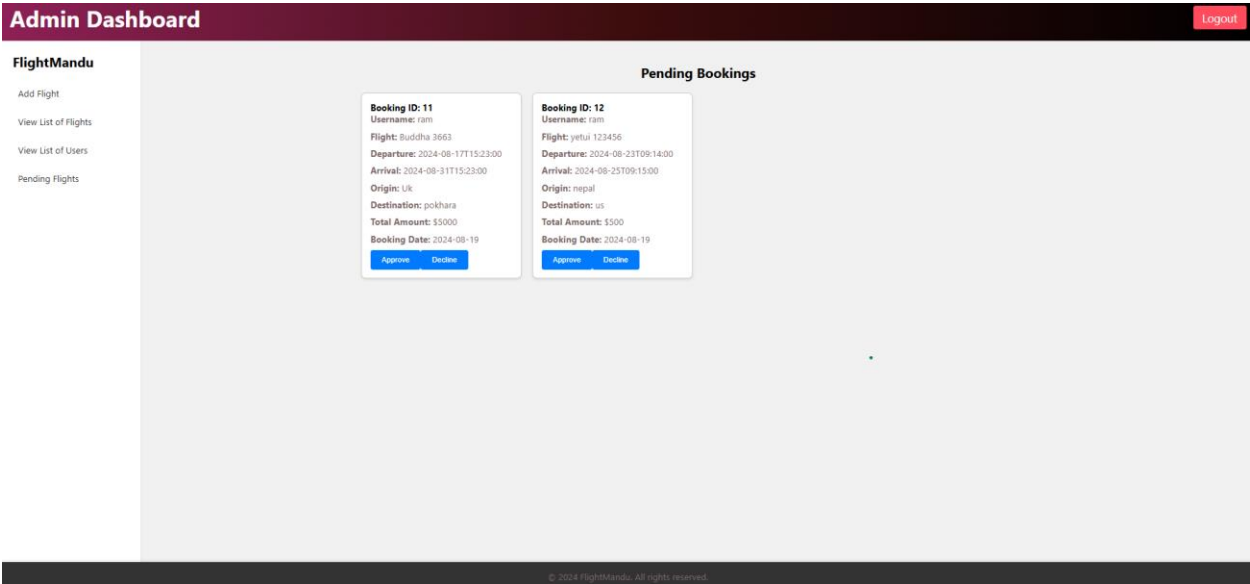


Figure 12Pending-flights

Tools and Technologies

React: A JavaScript library for building dynamic, component-based user interfaces.

Typescript: A superset of JavaScript that adds static typing for more reliable code.

Spring-Boot: A Java framework for quickly building production-ready backend services and APIs.

Postman: A tool for creating, testing, and managing API requests and responses.

IntelliJ IDEA: A powerful IDE for Java development with advanced coding and debugging features.

Visual Studio: A versatile IDE from Microsoft for developing applications in various languages, including .NET and C++.

Microsoft Word: A word processing application for creating and formatting text documents.



Figure 13ToolsNTech

ER Diagram

An ER diagram visually represents entities in a database and their relationships. Entities (e.g., User, Flight) are shown as rectangles, while attributes (e.g., UserID, Name) are ovals connected to their entities. Relationships (e.g., User books Flight) are depicted as diamonds connecting the entities. Primary keys uniquely identify entities, and foreign keys reference primary keys in related entities. Cardinality indicates the number of associations between entities, such as one User having many Bookings.

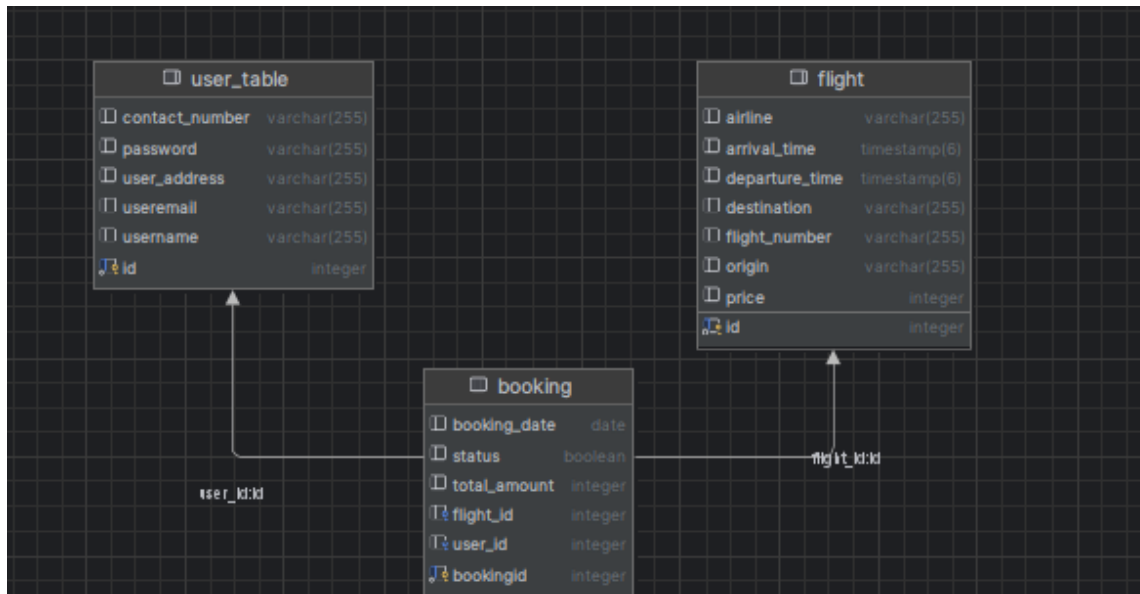


Figure 14Er diagram

Use-case Diagram

A use case diagram illustrates how users interact with a system and what functionalities the system provides. Actors (e.g., Customer, Admin) are represented as stick figures, while use cases (e.g., Book Flight, Manage Users) are shown as ovals. Relationships between actors and use cases, depicted by lines, show how actors interact with various system functionalities. System boundaries define the scope of the system, and associations indicate the interactions between actors and use cases. This diagram helps visualize user requirements and system functionality.

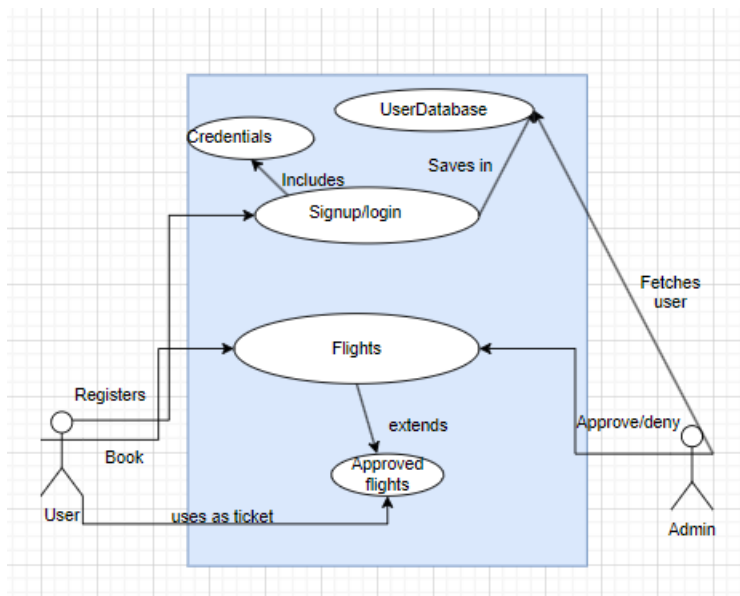


Figure 15 Use Case

Database Overview

In a database-driven application, the flow of operations starts when a user makes an HTTP request. The request is first handled by the Controller, which determines the appropriate action and forwards the request to the Service layer. The service processes the request, applying business logic and interacting with the Repository to access or modify data.

The Repository performs the actual database operations using Entities, which represent the structure of the database tables. These entities are used to retrieve or update data in the database. During this process, POJOs (Plain Old Java Objects) may be used to transfer data between different layers of the application, such as from the service to the controller.

Once the service completes the data operations, it returns the result to the controller. The controller then prepares and sends the final HTTP response back to the user. This structured flow ensures a clear separation of concerns, with each component responsible for specific tasks, leading to an organized and maintainable application.

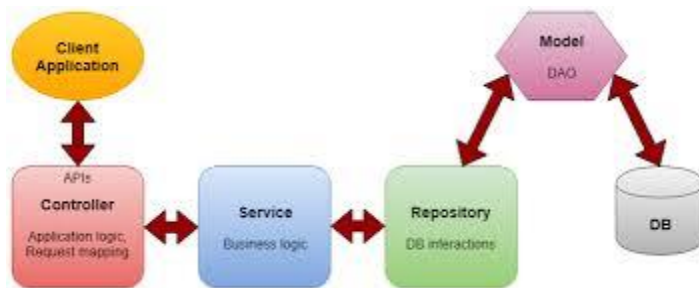


Figure 16Backend-Flow

Testing

Testing is the process of evaluating software to ensure that it functions correctly, meets specified requirements, and performs reliably under various conditions. It involves running tests to identify defects, verify that features work as intended, and ensure that the software behaves as expected in different scenarios.

Unit Testing involves testing individual components or functions of the software in isolation to confirm that each part performs its intended function correctly. This approach helps detect and fix bugs early in the development cycle, facilitates safe refactoring by ensuring that changes do not introduce new issues, and provides immediate feedback to developers about the correctness of their code. Tools like JUnit for Java and Jest for JavaScript are commonly used for writing and running unit tests.

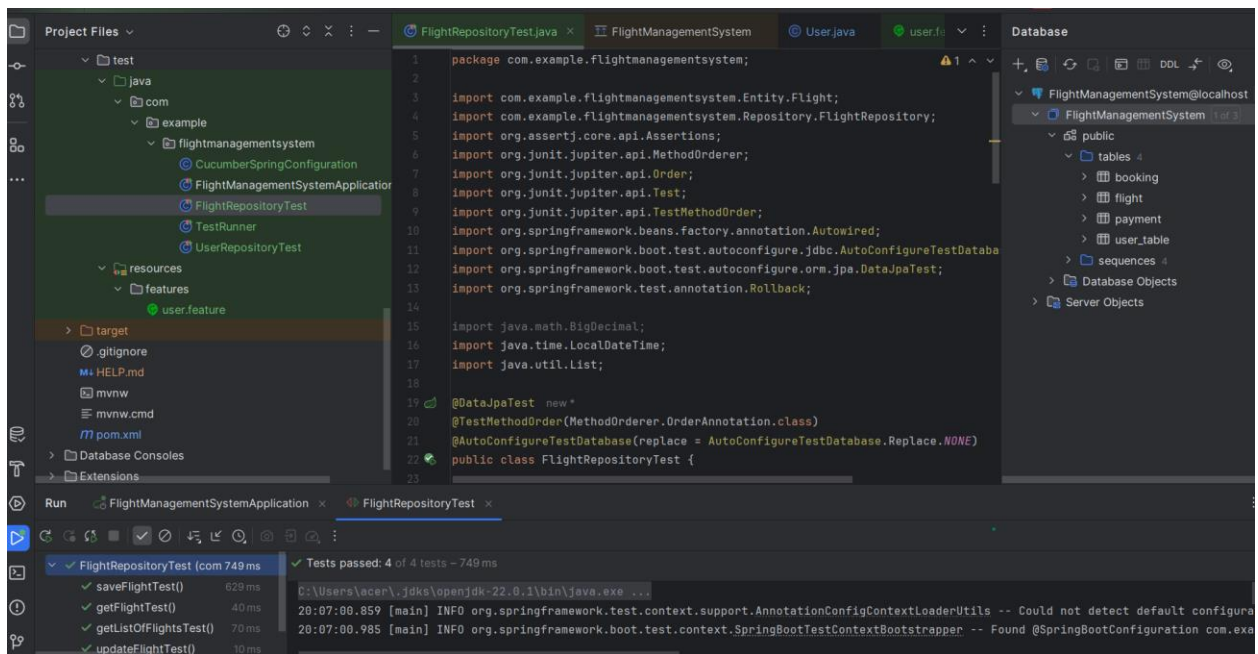


Figure 17UnitTest1

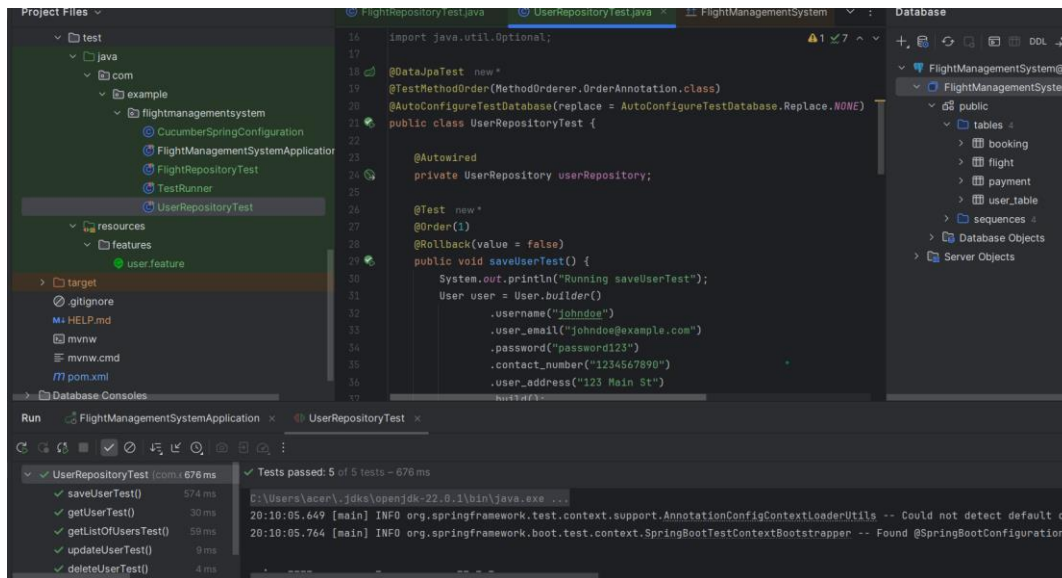


Figure 18UnitTest2

Behavior-Driven Development (BDD) is a testing methodology that emphasizes testing from the perspective of the end-user by describing the behavior of the application in natural, business-friendly language. BDD encourages collaboration between developers, testers, and stakeholders to define and refine feature requirements through scenarios that outline expected behavior. This approach ensures that the application meets business needs and user expectations. Tools such as Cucumber support BDD by allowing scenarios to be expressed in a readable format that bridges the gap between technical and non-technical team members.

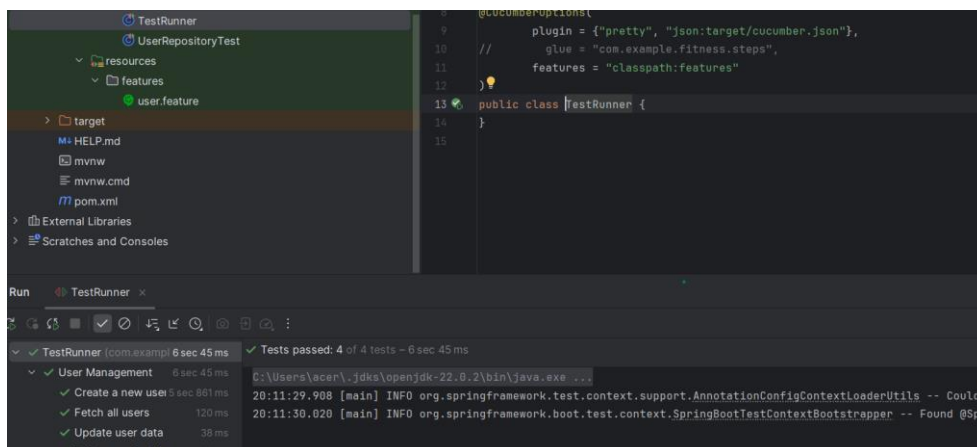


Figure 19BDDtest

Future features

- **Advanced Filtering and Sorting:** Enhance the search functionality with more granular filters and sorting options, such as by departure time, arrival time, or flight duration, to improve user experience and help users find the best flights more easily.
- **User Reviews and Ratings:** Implement a system for users to leave reviews and ratings for flights and airlines. This feature will provide valuable feedback for other users and help them make informed decisions.
- **Dynamic Pricing:** Integrate dynamic pricing algorithms that adjust flight prices based on factors such as demand, booking time, and availability, offering more competitive rates and maximizing revenue.
- **Integrated Payment Gateway:** Develop a secure payment gateway for seamless booking transactions, supporting multiple payment methods like credit/debit cards, digital wallets, and bank transfers.
- **Personalized Recommendations:** Use machine learning to offer personalized flight recommendations based on user preferences, previous searches, and booking history, enhancing the user experience.

Conclusion

Implementing advanced features will significantly enhance our application's functionality and user experience. Adding advanced filtering, dynamic pricing, and secure payment options will provide users with greater control and convenience during their booking process. Personalized recommendations and a mobile app will further tailor the experience to individual needs, while real-time flight tracking and multi-language support will improve accessibility and usability.

Incorporating user reviews and a loyalty program will foster user engagement and satisfaction, helping to build a more interactive and rewarding platform. Additionally, an advanced analytics dashboard will offer valuable insights for administrators, enabling data-driven decision-making.

These improvements are designed to address user needs, increase competitiveness, and ensure our application remains a leading solution in the market.

Reference